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Optimizing carbon capture and storage infrastructure including physics-based reservoir modelling

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ABSTRACT

The deployment of carbon capture and storage (CCS) requires a potentially complex infrastructure to transport and store CO_2 underground. Its optimal roll-out is key to limiting costs and enabling a timely deployment in line with ambitious mitigation scenarios. However, identifying the optimal design of the infrastructure is computationally challenging. Within this framework, mixed-integer linear programming (MILP) offers a computationally effective solution, which, however, requires linear models. Not surprisingly, geological reservoirs are typically represented as static sinks with constant injection rates and storage capacities as parameters. This approach neglects the dynamic properties of CO_2 injection, such as the reservoir pressure evolution over time, limiting the ability to evaluate their impact on the CCS chain design.

In this work, we propose a novel MILP model that integrates physics-based reservoir modelling into CCS chain optimization. Extending existing work on reduced-order modelling of reservoirs, we combine proper orthogonal decomposition and trajectory piecewise linearization to obtain a precise, yet computational efficient MILP model for the dynamic behaviour of CO_2 injection. Compared to full-scale reservoir simulations, the model computes the pressure around the injection well with ±5% accuracy and significant computational speed-ups (500-1800 times faster). We demonstrate its application in a full chain MILP model with an illustrative case study optimizing the decarbonization of a small industrial cluster through CCS, highlighting the model's ability to couple optimal operation of capture technologies with varying injection rates and to ensure the reservoir safety constraints while designing the chain.

1. Introduction

In one of his masterpieces, *The Last Supper*, Leonardo da Vinci developed a new fresco technique – applying tempera and oil to dry plaster – to allow finer detail and more painting time. Unfortunately, the fresco started to decay within a few decades. It was only thanks to numerous restoration efforts that we can still visit and admire the painting in Milan. Leonardo's new method was most likely carefully considered and tested, but did not account for a critical system behaviour: the material interactions over time in the presence of humidity. In modelling physical systems – whether walls painting, or underground reservoirs – it is often not what the model includes that leads to miscalculations, but what is quietly left out. This paper aims at decreasing the possibility of such oversights in the design of optimal carbon capture and storage supply chains: it introduces a physics-based reservoir model that can be directly embedded in large scale mixed-integer linear frameworks of CO_2 infrastructure.

Carbon capture and storage (CCS) is a decarbonization option where the CO_2 arising from the use of carbon-based fuels, most notably fossil fuels, is largely removed and prevented from entering the atmosphere by permanently storing, e.g. in geological formations. It is particularly important for some industrial sectors, where carbon-free options are not available, not scalable, or not cost competitive (IEA, 2023). In addition to the deployment of a suitable capturing process, CCS requires the development of an infrastructure for transporting and storing CO_2 (IEA, 2022). This development must be coordinated between different sources — typically industrial emitters — along with transportation modes and potential storage sites. The share of costs of building the transport and storage infrastructure for a CCS project can be substantial (Becattini et al., 2022). Moreover, the design of the whole CCS chain faces multiple complexities and must be co-optimized weighting different criteria including locations of emitters and storage reservoirs, cost, emissions, and environmental impact (Becattini et al.,

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Nomenclature

Variables

b	Reciprocal formation volume factor
C_{CO_2}	CO_2 fraction in injected flow
d	Distance
d_{ci}	Cumulative injection distance
d_{sat}	Relative saturation distance
f	Objective function
g	Gravitational constant
h	Vertical distance surface-to-well
k_r	Relative permeability
K	Intrinsic permeability
l	Total number of modes
l_p	Number of pressure modes
l_{sat}	Number of saturation modes
N_b	Number of gridblocks
N_u	Number of wells
p	Pressure
p_{bh}	Bottom-hole pressure
$p_{\max}$	Caprock fracture pressure
$p_{\text{pump, in/out}}$	Pump inlet/outlet pressure
p_{well}	Well cell pressure
p_{wh}	Wellhead pressure
S	Saturation
WI	Well index
W_{pump}	Pump power requirement
z	Reservoir depth
$\dot{m}$	Mass flow of CO_2
α	Fluid phase (water w, gas g)
η_{pump}	Pump efficiency
λ	Total mobility
μ	Viscosity
ν	Specific volume
ρ	Density
ϕ	Porosity
$\mathbf{A}$	Jacobian of res. eq. wrt previous state
$\mathbf{B}$	Jacobian of res. eq. wrt control variables
$\mathbf{C}$	Constraint matrix continuous var.
$\mathbf{D}$	Constraint matrix binary var.
$\mathbf{e}$	Constant MILP vector
$\mathbf{g}$	Residual equation
$\mathbf{J}$	Jacobian of res. eq. wrt current state
$\mathbf{u}$	Volumetric injection rate
$\mathbf{v}$	Continuous MILP variables
$\mathbf{w}$	Binary MILP variables
$\boldsymbol{\xi}$	Reduced states
$\boldsymbol{\upsilon}$	Darcy velocity
$\mathbf{x}$	States (pressure/saturation)
$\boldsymbol{\Phi}$	POD basis
$\boldsymbol{\Psi}$	Constraints reduction matrix

Indices

j	Timestep of a training run
n	Timestep of reservoir model
n'	Timestep CCS chain excluding storage

Sets

$\mathbb{R}$	Real numbers
S^n	Search range at timestep n
D^n	Distances at timestep n
T	6-month time horizon (reservoir)
T'	Daily time horizon

Acronyms

BHP	Bottom-hole pressure
CAPEX	Capital cost
CC	Carbon capture
CCS	Carbon capture and storage
MILP	Mixed-integer linear programming
MRST	MATLAB Reservoir Simulation Tool
POD	Proper orthogonal decomposition
PCC	Post-combustion capture
ROM	Reduced-order model
SVD	Singular value decomposition
TPWL	Trajectory piecewise linearization
WtE	Waste to energy plant

2022; Roussanaly et al., 2021; Burger et al., 2024). Therefore, to enable the timely and economical deployment of CCS, it is essential to develop tools capable of accurately modelling and optimizing the CCS chain. As these tools need to be temporally and spatially resolved, which leads to several thousands of optimization variable, models need to be linear and simple.

Consequently, this issue has been tackled mostly with linear or mixed-integer linear programming (MILP) models (Middleton and Bielicki, 2009; Becattini et al., 2022; d'Amore and Bezzo, 2017; Jones et al., 2022; Middleton et al., 2012; Aras et al., 2025). These models typically represent the elements of the CCS chain using simple, linear input–output relationships to keep the optimization problem computationally tractable. This approach allows to model complex CCS networks with multi-year time horizons while retaining hourly/daily resolution and while including e.g. different transportation modes or geographical features (Becattini et al., 2022; Crîstiu et al., 2023). However, coming back to Leonardo, this approach naturally limits how accurately the model can represent the physical reality of the CCS chain components, e.g. non-linear behaviours of compressors, pumps, pipelines and reservoirs.

In particular, in MILP problems, CO_2 storage in geological formations (mainly saline aquifers and depleted oil and gas fields) is typically modelled as a static sink where CO_2 can be injected at a certain rate. However, the injection of CO_2 in underground reservoirs is a dynamic process and important properties of the system, such as reservoir pressure, plume location and injectivity, change over time and depending on the site operation (Celia and Nordbotten, 2012). These factors can have an impact on the design of the CCS chain in different ways. For example, a change in wellhead pressure affects the CO_2 transport capacity of the pipeline connecting the compression station with the storage site (Anon, 2019). The evolution of the wellhead pressure also impacts downstream pressure, sizing and energy consumption of the compression/pumping system. At the same time, different operating pressures of the pipelines correspond to different line packing times (Martynov et al., 2024), which can affect the system's flexibility. Additionally, over the lifetime of a CCS project, the injection of CO_2 should be such that the pressure in the reservoir never exceeds the fracture pressure of the caprock to ensure the integrity of the reservoir (Ringrose, 2020). Therefore, understanding and including the dynamics of the reservoir is important when designing the CCS chain.

However, modelling the dynamics of injection or extraction of fluids in a porous reservoir (either for CO_2 storage or hydrocarbon production) is a highly computationally demanding task. In particular, it requires solving non-linear partial differential equations discretized both in time and space (such as the multiphase Darcy flow equations), which is done by adopting finely meshed grids, i.e. introducing hundreds of thousands of discretization points (Celia and Nordbotten, 2012). Although these models have been applied to certain optimization problems (Güyagüler, 2002; Temizel et al., 2017), their use in large optimization problems is computationally prohibitive. Still, the potential benefits (or, the requirement) of optimizing reservoir operations are significant. Therefore, in contexts other than CCS chain optimization with MILP, many efforts have been made to develop more efficient models that capture the behaviour of the reservoir.

For example, Santibanez-Borda et al. (2019) developed a surrogate model to map the evolution of the pressure increase in a reservoir based on combinations of CO_2 injection and brine extraction profiles. To do so, they used (regularized) linear regression and multivariate adaptive regression splines to fit the results of 100 full-scale simulations. With this model, they optimized the operations of the injection and extraction wells to maximize the storage capacity CO_2 of the fields. Cardoso and Durlofsky (2010) proposed a reduced-order model (ROM) based on a combination of trajectory piecewise linearization (TPWL) and proper orthogonal decomposition (POD). With this model, they compute the new states of the system using a linear expansion around previously simulated states (obtained from training runs of high-fidelity, computationally expensive simulations). The states are represented in a reduced space obtained through the POD, which projects the initial space onto a lower-dimensional one, minimizing the loss of information. The combination of these two techniques allowed the development of a computationally efficient model, which they used to optimize the cumulative production of an oil field. The work by Cardoso and Durlofsky served as a starting point for multiple research streams (He and Durlofsky, 2014; Jin and Durlofsky, 2018). In particular, Jin and Durlofsky (2018) developed a POD-TPWL model of the rate-based injection of CO_2 in a saline aquifer. With this model, they minimized the amount of CO_2 flowing to a layer of the reservoir associated with the risk of CO_2 leakage. Coutinho et al. (2021) used the POD-TPWL formulation to propose a physics-based deep learning ROM. Similarly, the discrete empirical interpolation method (DEIM) is another technique for reservoirs ROM that uses the idea of projecting the original high-dimensional state space onto a lower one, and that has been combined with TPWL and POD (Tan et al., 2019; Chaturantabut and Sorensen, 2011).

Overall, multiple simplified reservoir models have been developed for various types of optimizations. However, there is a clear gap in the open literature: to the best of our knowledge, no work bridges the modelling of CO_2 injection in porous reservoirs with the optimization of the entire CCS chain; indeed, a physics-based, computationally efficient MILP formulation of CO_2 storage is missing. Building upon the POD-TPWL method for rate-controlled wells detailed by Jin and Durlofsky (2018), this paper contributes at closing this gap by:

- Developing an MILP version of the POD-TPWL method for use in large scale optimization problems (e.g. energy systems, CO_2 chains).
- Enhancing the physical fidelity of CCS infrastructure optimization by including a physics-based model of CO_2 reservoirs.
- Validating the MILP model against the full scale 3D model and discussing the main parameters and procedures that need specific attention.
- Demonstrating the use of the MILP model in an exemplary CCS chain optimization.

The paper is structured as following. First, we elaborate on how to develop the linear model in the general context of the MILP framework: We start from full-scale reservoir simulations, explaining the POD-TPWL procedure with the adjustments required and outlining its MILP formulation. Then, we show the model's performance across different injection scenarios and its potential use in a simple case study optimizing a CCS chain over a 5-year period. Lastly, we discuss the limitations and other potential uses of the model.

2. Methods: An MILP model for CO_2 storage

In an MILP problem, the aim is to minimize an objective function $f(\mathbf{v}, \mathbf{w})$ – such as the system's total cost or emissions – while ensuring that a certain set of constraints are not violated. The objective function and constraints are linear, and the variables are real or integer numbers (the latter, typically reformulated as binaries). The problem can be formulated as follows:

$$\begin{aligned} \min_{\mathbf{v},\mathbf{w}} \quad & f(\mathbf{v}, \mathbf{w}) \\ \text{s.t.} \quad & \mathbf{C}\mathbf{v} + \mathbf{D}\mathbf{w} = \mathbf{e}, \\ & \mathbf{v} \geq 0, \quad \mathbf{v} \in \mathbb{R}^{N_v}, \\ & \mathbf{w} \in \{0,1\}^{N_w}. \end{aligned} \tag{1}$$

where $\mathbf{v}$ and $\mathbf{w}$ represent the continuous and binary variables. The matrices $\mathbf{C}$ and $\mathbf{D}$ define the associated constraints, with $\mathbf{e}$ being the known term of the constraints. Therefore, any specific physical model is included in the MILP via subsets of the constraints $\mathbf{C}$, $\mathbf{D}$ and $\mathbf{e}$.

Fig. 1 shows the steps required to develop a reservoir model suitable for an MILP optimization problem. The starting point is the availability of simulations from a trusted full-scale reservoir simulation model. These can then be provided as input to the POD-TPWL method, which produces as output a reduced-order model of the reservoir. The latter needs to be fully linearized and formulated as constraints in an MILP problem. Lastly, the MILP model needs to be embedded in a CO_2 chain optimization framework, where it can complement other set of constraints, e.g. for the capture technologies and transport network. Let us go through the different steps in more detail in the following.

2.1. Full-scale reservoir simulation

The first step to obtain a linear model for CO_2 storage using the POD-TPWL approach is to derive simulations from full-scale reservoir simulations; these serve as training set of the POD-TPWL model. In this work, we obtain the required simulations using the open-source software MATLAB Reservoir Simulation Toolbox (Lie, 2016) (MRST). The simulations are based on compressible, immiscible two-phase flow, and the backbone equations of the system are represented by the equation of conservation of mass for each phase (Eq. (2)) and the multiphase Darcy equation for the phase velocity (Lie, 2016) (Eq. (3)).

$$\rho_\alpha^{\text{surf}} \left(\frac{\partial}{\partial t}(S_\alpha \phi b_\alpha) + \nabla \cdot (b_\alpha \boldsymbol{v}_\alpha) \right) = \rho_\alpha^{\text{surf}} u_\alpha, \quad \alpha \in w, g \tag{2}$$

Here, u_α is the injection/extraction rate of the phase α (either gas g or water w), $\boldsymbol{v}_\alpha$ and S_α are respectively the Darcy velocity and the saturation of the corresponding phase, with brine and gaseous/supercritical CO_2 making up the pore volume ϕ through the saturation constraint $S_w + S_g = 1$. The equations are here given in surface volumes through reciprocal formation volume factors b_α that relate the volume at reservoir conditions to that of the given surface conditions with surface density $\rho_\alpha^{\text{surf}}$. For the immiscible system in particular, we have that the phase density can be expressed by a simple scaling by the density of the phase at surface conditions, $\rho_\alpha = b_\alpha \rho_\alpha^{\text{surf}}$. The phase fluxes at reservoir conditions are given by a standard multiphase extension of Darcy's law:

$$\boldsymbol{v}_\alpha = -\lambda_\alpha K (\nabla p_\alpha - \rho_\alpha g \nabla z), \quad \alpha \in w, g \tag{3}$$

where K is the intrinsic permeability of the rock, p_α the phase pressure, $\rho_\alpha g \nabla z$ the gravitational potential gradient and $\lambda_\alpha = k_{r\alpha}(S_\alpha)/\mu_\alpha$ the mobility - a ratio of the relative permeability and phase viscosity.

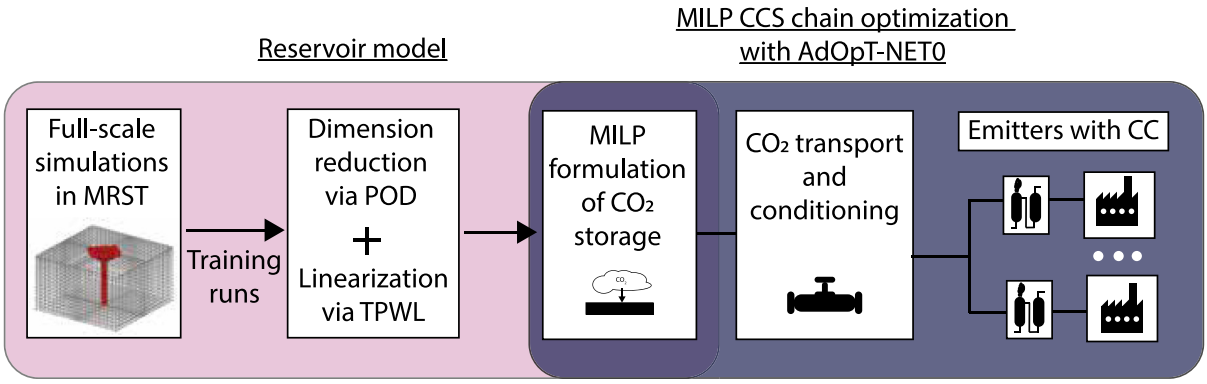


Fig. 1. Methodological framework of the paper.

The system has four variables (pressure and saturation of the gas and water phase respectively), but only two equations. To complete the system, the saturation equation constraint and the relation between phase pressures and the capillary pressure $p_g = p_w + p_c(S_g)$ need to be added. After reducing the variable set by these two additional equations, the system can be solved using two primary variables, which in this work are the saturation of the CO_2 phase and the pressure. To simulate the injection of CO_2, the reservoir and the equations are discretized with a finite volume formulation with a 3-dimensional grid with N_b grid blocks. For every grid block, the Eqs. (2) and (3) must be solved. Therefore, the overall system has $2N_b$ variables. Using a fully implicit time discretization, the system can be written as

$$\mathbf{g}\left(\mathbf{x}^{n+1}, \mathbf{x}^{n}, \mathbf{u}^{n+1}\right) = \mathbf{0} \tag{4}$$

where $\mathbf{g} \in \mathbb{R}^{2N_b}$ is the residual equation that needs to be minimized to zero (typically via the Newton method); $\mathbf{x}^n \in \mathbb{R}^{2N_b}$, $\mathbf{x}^{n+1} \in \mathbb{R}^{2N_b}$ are the vectors of the states (pressure and saturation in each grid block) at timestep n and $n+1$ respectively, $\mathbf{u}^{n+1} \in \mathbb{R}^{N_u}$ is the control parameter vector of the N_u wells. This can be the injection rate (or potentially the production rate of brine from another well) or the target bottom-hole pressure (BHP) for the injection. In the context of this work, we chose as control parameter in the vector $\mathbf{u}$ the injection rate, as it allows us to treat the injection rate as an optimization variable of the MILP problem (i.e. the injection rate is determined by the minimization of the objective function $f(\mathbf{v}, \mathbf{w})$). Different control parameters would also be possible and need to be implemented depending on the formulation of the MILP.

In every timestep n, we aim at solving the system represented in Eq. (4) to find the new state $\mathbf{x}^{n+1}$, knowing the current state $\mathbf{x}^n$ and the control value $\mathbf{u}^{n+1}$ in the next timestep (e.g. the injection rate of CO_2). However, this system is nonlinear and its solution via the Newton method solution requires, for every timestep, multiple assemblies and solutions of both the residual function and the corresponding Jacobian matrix of the system, which for iteration k can be written:

$$\mathbf{x}_{k+1}^{n+1} = \mathbf{x}_{k}^{n+1} + \Delta\mathbf{x}_k \qquad \Delta\mathbf{x}_k = -\left(\frac{\partial \mathbf{g}\left(\mathbf{x}^{n}, \mathbf{x}^{n-1}, \mathbf{u}^{n}\right)}{\partial \mathbf{x}^{n}}\right)^{-1} \mathbf{g}\left(\mathbf{x}^{n+1}, \mathbf{x}^{n}, \mathbf{u}^{n+1}\right). \tag{5}$$

This process is repeated until the norm of the residual equation is sufficiently small. The Jacobian matrix has dimensions $2N_b \times 2N_b$ where the number of grid cells N_b can be anywhere from a few thousand cells to several millions.

2.2. Reduced-order model

Clearly, the complexity of solving this system of equation prevents using it directly in the context of large optimization problems, as for example an MILP of the optimal CCS infrastructure. The development of a linearized POD-TPWL with an MILP formulation tackles these problems by (1) linearizing the governing equations and shifting the computation of the Jacobian to the preprocessing steps instead of the online ones, (2) reducing the number of variables via POD, and (3) developing the necessary linear constraints to run the model in an MILP framework.

2.2.1. Linearization via trajectory piecewise linearization

In this and the following sections, we briefly describe the POD-TPWL procedure developed by Jin and Durlofsky (2018); we do that by highlighting the features and changes required to develop the MILP version of the procedure. The interested reader can refer to the original work for additional details.

The first step of the POD-TPWL method consists of conducting multiple full-scale simulations of CO_2 injection into the reservoir; these simulations will serve as training simulations of the reduced-order model. For each simulation, we set a CO_2 injection profile over time; this aligns with having the injection rate as a control parameter of the full scale model and should be adapted whenever another approach is used, e.g. bottom-hole pressure fixed. As we discuss later, the selection of injection rates should represent reasonable values for the specific modelling problem. Throughout these simulations, we record the states (e.g. pressure and saturation in each grid block) and the system's derivatives at every time step: the core concept of the linearization part (TPWL) is to use these recorded states and derivatives to facilitate new simulations. Specifically, when running a new (linearized) simulation, the new states are computed as linear approximations of the governing equations, linearized around the point that is closest to the current state among the states recorded during the training runs. This approach uses the stored data to approximate the behaviour of the system under new conditions, thereby significantly reducing computational costs while maintaining a high level of accuracy in the simulations, especially when the new conditions are reasonably close to the training ones.

The linear approximation of the residual equation in the new simulation (where the index n represents the timestep of the new simulation, as opposed to the index i representing that of the training runs) is performed using a first-order Taylor expansion of the residual equation around a training solution. Given the solution of the new simulation at timestep n and a solution $\left(\mathbf{x}^{i+1}, \mathbf{x}^{i}, \mathbf{u}^{i+1}\right)$ of the residual equation from the training run, we have that:

$$\begin{aligned}\mathbf{g}^{n+1} \approx \mathbf{g}^{i+1} &+ \frac{\partial \mathbf{g}^{i+1}}{\partial \mathbf{x}^{i+1}}(\mathbf{x}^{n+1} - \mathbf{x}^{i+1}) \\ &+ \frac{\partial \mathbf{g}^{i+1}}{\partial \mathbf{x}^{i}}(\mathbf{x}^{n} - \mathbf{x}^{i}) + \frac{\partial \mathbf{g}^{i+1}}{\partial \mathbf{u}^{i+1}}(\mathbf{u}^{n+1} - \mathbf{u}^{i+1}) = 0\end{aligned} \tag{6}$$

where we defined $\mathbf{g}^{n+1} = \mathbf{g}(\mathbf{x}^{n+1}, \mathbf{x}^{n}, \mathbf{u}^{n+1})$ and $\mathbf{g}^{i+1} = \mathbf{g}(\mathbf{x}^{i+1}, \mathbf{x}^{i}, \mathbf{u}^{i+1})$. By definition of $\mathbf{g}$, and knowing that i refers to a training run (i.e. its residual has already been driven to 0), we also have that $\mathbf{g}^{i+1} = 0$. Rearranging Eq. (6) and introducing some new notation we have

$$\mathbf{J}^{i+1}\left(\boldsymbol{x}^{n+1} - \boldsymbol{x}^{i+1}\right) = -\left[\mathbf{A}^{i+1}\left(\boldsymbol{x}^{n} - \boldsymbol{x}^{i}\right) + \mathbf{B}^{i+1}\left(\boldsymbol{u}^{n+1} - \boldsymbol{u}^{i+1}\right)\right] \tag{7}$$

where we defined $\mathbf{J}^{i+1} = \frac{\partial \boldsymbol{g}^{i+1}}{\partial \boldsymbol{x}^{i+1}} \in \mathbb{R}^{2N_b \times 2N_b}$, $\mathbf{A}^{i+1} = \frac{\partial \boldsymbol{g}^{i+1}}{\partial \boldsymbol{x}^{i}} \in \mathbb{R}^{2N_b \times 2N_b}$, and $\mathbf{B}^{i+1} = \frac{\partial \boldsymbol{g}^{i+1}}{\partial \boldsymbol{u}^{i+1}} \in \mathbb{R}^{2N_b \times N_u}$ for the Jacobian matrices of the residual function with respect to $\boldsymbol{x}^{i+1}$, $\boldsymbol{x}^{i}$, and $\boldsymbol{u}^{i+1}$, respectively.

It is important to note that obtaining these Jacobians from full reservoir simulations might be fairly complex, and it needs to be done while having the underlying logic of the model and the control parameters clear. For example, when adopting the MRST as full scale reservoir model along with the injection rate as control parameter (as the case in this work), it is not possible to directly use the Jacobians $\mathbf{J}^{i+1}$ and $\mathbf{A}^{i+1}$; this would be possible only in the case of wells controlled by the BHP directly. The reason is that, in MRST framework, the wells are modelled by a separate set of governing equations for surface rates and BHP: the injection rate in the well cells is obtained from the BHP through the inflow-performance relation (Lie, 2016). We provide a detailed description of how to use the Jacobians from MRST in the TPWL with injection rate as control parameter in the Supplementary Information.

Next to the linearization process and the identification of the Jacobians, a key feature of TPWL is the dynamic selection of the point around which the linearization is performed at every timestep n. Unlike a fixed-point linearization, TPWL selects this point based on its proximity to the current state, thus providing a more accurate trajectory of the evolution of the system. Although multiple simulations are necessary for the POD (see below), we use only one training run for the point selection. Multiple training runs for the point selection would improve the accuracy of the model (Jin and Durlofsky, 2018), but at the expenses of larger computational time during the online simulation, which must be kept to the minimum in the context of CCS chain MILP optimization.

As proposed by Jin and Durlofsky (2018), the identification of the closest point can be executed as follows. For every timestep n, the distance between the current state n and a series of states from the training run is calculated. Given $s \in \mathbb{N}$ as the range of search, the training states identified are $j \in S^n$ with $S^n = [n-s, n-s+1, \dots, n+s-1, n+s]$. In this work, we adopt s=1. While choosing a larger value might improve the ability of the model to capture large deviations from the training simulations, it would also substantially increase the computational complexity, which must be constrained given the scope of the model. Moreover, as demonstrated in the Results section, this choice does not prevent the model from achieving a good predictive accuracy, rendering its impact on the optimization problem negligible. The distance is defined as the sum of the relative difference in saturation $d_{\text{sat}}^{n,j}$ and the one in cumulative CO_2 injected $d_{\text{ci}}^{n,j}$.

$$d^{n,j} = d_{\text{sat}}^{n,j} + \gamma d_{\text{ci}}^{n,j} \tag{8}$$

with γ as a weighting factor. The relative saturation distance is

$$d_{\text{sat}}^{n,j} = \frac{|\xi_{\text{sat}}^{n} - \xi_{\text{sat}}^{j}|}{|\xi_{\text{sat}}^{n}| + \varepsilon} \tag{9}$$

and the distance in cumulative CO_2 injected is

$$d_{\text{ci}}^{n,j} = \sum_{k=1}^{N_u} \frac{\left| \int_0^{t^n} u_k^n \mathrm{d}t - \int_0^{t^j} u_k^j \mathrm{d}t \right|}{\int_0^{t^n} u_k^n \mathrm{d}t + \varepsilon} \tag{10}$$

The parameter $\epsilon = 0.001$ is added to prevent division by very small numbers. The computation of the distance returns a set D^n of $2s+1$ distances for every timestep n, where a minimization is performed to obtain the closest training state $j \in S^n$. We note here that the distance computed with Eq. (8)–(10) cannot be used as it is in an MILP framework, and therefore it should be appropriately modified. We come back to this point with the formulation of the MILP model.

2.2.2. Dimension reduction via proper orthogonal decomposition

With POD, the objective is to find an efficient way to represent the system – here the evolution of pressure and saturation over time in each grid block – using fewer dimensions while minimizing the loss of information. To achieve this, POD constructs a new orthogonal basis matrix $\boldsymbol{\Phi} \in \mathbb{R}^{2N_b \times l}$ that spans a lower-dimensional subspace of dimension l. The system states can then be approximated by projecting them onto this subspace as:

$$\mathbf{x} \approx \boldsymbol{\Phi}\boldsymbol{\xi} \tag{11}$$

where $\boldsymbol{\xi} \in \mathbb{R}^l$ is the reduced representation of the states. To determine the POD basis matrix, we first gather a series of snapshots from the full-scale training simulations, recording the system states at various time points across multiple simulations. In this process, we separate the pressure and the saturation, creating two snapshot matrices of dimension $\mathbb{R}^{N_b \times L}$ and $\boldsymbol{\Phi}_{\text{sat}} \in \mathbb{R}^{2N_b \times L}$ each, where each column represent the states (pressure and CO_2 saturation, respectively) at an instant of time and L is the total number of snapshots saved. Once these snapshots are collected, we independently perform a singular value decomposition (SVD) on the two snapshot matrices. In this way, we can take the left singular vectors associated with the highest energy (i.e. associated eigenvalue of the correlation matrix) for each SVD (in number l_p and l_{sat} for the pressure and saturation snapshots respectively) and create the orthonormal basis matrices $\boldsymbol{\Phi}_p$ and $\boldsymbol{\Phi}_{\text{sat}}$, which have the left singular vectors as columns. The choice of how many singular vectors are kept, i.e. the value of l_p and l_{sat}, depends on the specific problem; often, a trial and error procedure is necessary to find a suitable value (i.e. trying different l_p and l_{sat} until the result is satisfactory). It is worth noting that, since in the optimization of the CCS chain we are interested more in the pressure evolution in the reservoir rather than the plume location, we discard all the modes associated with the saturation variables except the one with the highest energy, i.e. we choose $l_{\text{sat}} = 1$. This decision further reduces the computational requirements of the model while preserving the accuracy: we tested the model performance with different l_{sat} values, finding that their impact on the pressure prediction is negligible.

Once $\boldsymbol{\Phi}_p$ and $\boldsymbol{\Phi}_{\text{sat}}$ are obtained, we can write the states of the system (pressure $\boldsymbol{x}_p$ and saturation $\boldsymbol{x}_{\text{sat}}$) in terms of the corresponding reduced states $\boldsymbol{\xi}_p \in \mathbb{R}^{l_p}$ and $\boldsymbol{\xi}_{\text{sat}} \in \mathbb{R}^{l_{\text{sat}}}$ as:

$$\mathbf{x} = \begin{bmatrix} \mathbf{x}_p \\ \mathbf{x}_{\text{sat}} \end{bmatrix} \approx \boldsymbol{\Phi}\boldsymbol{\xi} = \begin{bmatrix} \boldsymbol{\Phi}_p & 0 \\ 0 & \boldsymbol{\Phi}_{\text{sat}} \end{bmatrix} \begin{bmatrix} \boldsymbol{\xi}_p \\ \boldsymbol{\xi}_{\text{sat}} \end{bmatrix} \tag{12}$$

After the projection of the states onto the subspace defined by the new basis $\boldsymbol{\Phi}$, the dimensions of the system are reduced from the initial $2N_b$ to $l = l_p + l_{\text{sat}}$, with l being much smaller than $2N_b$. By inserting the reduced states representation (Eq. (12)) in the linearized residual equation (Eq. (7)), we obtain a system with $2N_b$ equations and l unknowns, which means it is overdetermined. Therefore, we need to project this system onto an l-dimensional subspace to solve it. Being $\boldsymbol{\Psi} \in \mathbb{R}^{2N_b \times l}$ a basis of this subspace, we can multiply both sides of the new system by $\boldsymbol{\Psi}^T$ to obtain a solvable system:

$$\left(\boldsymbol{\Psi}^{i+1}\right)^T \mathbf{J}^{i+1}\boldsymbol{\Phi}\left(\xi^{n+1} - \xi^{i+1}\right) = -\left(\boldsymbol{\Psi}^{i+1}\right)^T \left[\mathbf{A}^{i+1}\boldsymbol{\Phi}\left(\xi^{n} - \xi^{i}\right) + \mathbf{B}^{i+1}\left(\boldsymbol{u}^{n+1} - \boldsymbol{u}^{i+1}\right)\right] \tag{13}$$

The matrix $\boldsymbol{\Psi}$ is not unique (He and Durlofsky, 2015) and therefore a choice on the type of projection should be made. One option is the Galerkin projection, in which $\boldsymbol{\Psi} = \boldsymbol{\Phi}$, which has been used in multiple studies (Cardoso and Durlofsky, 2010; He et al., 2013). However, the Petrov–Galerkin projection ($\boldsymbol{\Psi}^{i+1} = \mathbf{J}^{i+1}\boldsymbol{\Phi}$) has been shown to provide more stable results (Carlberg et al., 2011; He and Durlofsky, 2015) and it is therefore applied here. After the application of the projection, we introduce the following notation

$$\mathbf{J}_r^{i+1} = \left(\boldsymbol{\Psi}^{i+1}\right)^T \mathbf{J}^{i+1}\boldsymbol{\Phi}, \quad \mathbf{A}_r^{i+1} = \left(\boldsymbol{\Psi}^{i+1}\right)^T \mathbf{A}^{i+1}\boldsymbol{\Phi}, \quad \mathbf{B}_r^{i+1} = \left(\boldsymbol{\Psi}^{i+1}\right)^T \mathbf{B}^{i+1}$$

with $\mathbf{J}_r^{i+1} \in \mathbb{R}^{l \times l}$, $\mathbf{A}_r^{i+1} \in \mathbb{R}^{l \times l}$ and $\mathbf{B}_r^{i+1} \in \mathbb{R}^{l \times N_u}$. As a result, we can write the linearized residual equation in the reduced space as

$$\xi^{n+1} = \xi^{i+1} - (\mathbf{J}_r^{i+1})^{-1}[\mathbf{A}_r^{i+1}(\xi^{n} - \xi^{i}) + \mathbf{B}_r^{i+1}(\mathbf{u}^{n+1} - \mathbf{u}^{i+1})] \tag{14}$$

We can therefore obtain the evolution of the reduced states over time by solving Eq. (14) for any timestep n. From the reduced states, we can change the basis back to the original representation of the states, thereby providing a representation of the reservoir physics within a linear optimization framework, i.e. the CCS chain. It is worth stressing that after the reconstruction of the solution in the original space, the mass conservation equations may not be strictly satisfied; however, the resulting approximation error is considered acceptable for the purposes of this study.

2.3. MILP formulation of the POD-TPWL model and integration in the full chain framework

The development of the MILP formulation of the reservoir model involves several key steps. This section provides a detailed description of each step, while a flow diagram illustrating the overall process is included in the Supplementary Information. The core of the linear POD-TPWL model is the solution of the linearized residual equation in the reduced space (Eq. (14)). As we have formulated it, this equation returns the reservoir pressure at the timestep $n+1$ as a linear function of the two optimization variables: the injection rate u^{n+1} and the reservoir states in the previous timestep ξ^n. Implementing this equation in an MILP environment is rather straightforward.

However, the MILP model requires additional constraints, most notably for selecting the closest state i in the training simulation. The distance method highlighted above is a non-linear process because (i) the distance calculation (Eq. (9) and (10)) requires the absolute value of MILP variables, (ii) it normalizes the distances using values from the new simulation (which are variables of the MILP) and (iii) it involves the identification and selection of the minimum distance among the set D^n. Therefore, appropriate constraints and modifications need to be applied to make these steps compatible with an MILP formulation.

To ensure that the MILP reservoir model is computationally tractable and fully linear, we adopt a different approach to calculate the distance compared to previous works (Jin and Durlofsky, 2018; He and Durlofsky, 2014). Firstly, we neglect the component of the distance between each state in terms of the relative saturation difference per each mode (Eq. (9)). The reasons are that (i) we use only one saturation mode ($l_{\text{sat}} = 1$) and (ii) it would be extremely expensive to compute this term for more modes in an MILP framework: it entails the calculation of l_{sat} absolute values per each state in S^n, which, as we show below, requires extra auxiliary and integer variables. As a consequence, we have that the weighting term γ in Eq. (8) is not needed anymore and $d^{n,j} = d^{n,j}_{\text{ci}}$ Secondly, we modify the definition of $d^{n,j}_{\text{ci}}$ by normalizing the differences with the values from the training run instead of the new simulation. This is a necessary step to ensure that the entire model is linear.

Once the distances are calculated, we need to select the index j corresponding to the minimum distance in the set D^n, which is also the index i used to solve Eq. (14).

In terms of constraints, we first need to implement the absolute value from Eq. (10); this requires the definition of two auxiliary variables $d^{n,j}_{\text{ci,pos}}$ and $d^{n,j}_{\text{ci,neg}}$ such that

$$d^{n,j}_{\text{ci}} = d^{n,j}_{\text{ci,pos}} + d^{n,j}_{\text{ci,neg}} \qquad \forall \quad n \in T, j \in S^n \tag{15}$$

where T is the set of timesteps of the reservoir simulation (see next section for details about the time discretization). Then, we introduce a binary variable $a^{n,j} \in \{0,1\}$ and the following constraints:

$$d^{n,j}_{\text{ci,pos}} = \frac{\int_0^{t^n} u^n \mathrm{d}t - \int_0^{t^j} u^j \mathrm{d}t}{\int_0^{t^j} u^j \mathrm{d}t + \varepsilon} a^{n,j} \qquad \forall \quad n \in T, j \in S^n \tag{16}$$

$$d^{n,j}_{\text{ci,neg}} = \frac{-\left(\int_0^{t^n} u^n \mathrm{d}t - \int_0^{t^j} u^j \mathrm{d}t\right)}{\int_0^{t^j} u^j \mathrm{d}t + \varepsilon} \left(1 - a^{n,j}\right) \qquad \forall \quad n \in T, j \in S^n \tag{17}$$

where we are dropping the sum over the number of wells from the distance calculation because in this work we consider only a single injection well. We also note that in Eqs. (16) and (17) there is a bilinearity that can be removed using a big-M constraint. The absolute value in Eq. (10) is finally enforced by adding the constraints

$$d^{n,j}_{\text{ci,pos}} \geq 0 \qquad \forall \quad n \in T, j \in S^n \tag{18}$$

$$d^{n,j}_{\text{ci,neg}} \geq 0 \qquad \forall \quad n \in T, j \in S^n \tag{19}$$

In addition to absolute value, we need to add constraints to find the minimum distance in the set D^n. We therefore define another auxiliary variable $d^n_{\text{min}} \geq 0$ such that

$$d^n_{\text{min}} \leq d^{n,j}_{\text{ci}} \qquad \forall \quad n \in T, j \in S^n \tag{20}$$

$$d^n_{\text{min}} \geq \sum_{j \in S^n} q^{n,j} d^{n,j}_{\text{ci}} \qquad \forall \quad n \in T \tag{21}$$

where we introduced the binary variable $q^{n,j} \in \{0,1\}$, which must satisfy the following constraint to ensure that only one distance is selected

$$\sum_{j \in S^n} q^{n,j} = 1 \qquad \forall \quad n \in T \tag{22}$$

With constraints (20), we are forcing d^n_{min} to be smaller or equal than all distances in D^n. At the same time, constraint (21) states that d^n_{min} should also be bigger or equal to the same distances, times the binary variable $q^{n,j}$. If only one $q^{n,j}$ should be equal to one (constraint (22)), then these constraints are satisfied only when the index j corresponds to the minimum distance in the set D^n. We can then use $q^{n,j}$ to obtain the relevant parameters of the chosen training state to be used in Eq. (14) as

$$\xi^{i+1} = \xi^{i+1,n} = \sum_{j \in S_n} \xi^{j+1} \, q^{n,j} \qquad \forall \quad n \in T \tag{23}$$

$$\xi^{i} = \xi^{i,n} = \sum_{j \in S^n} \xi^{j} \, q^{n,j} \qquad \forall \quad n \in T \tag{24}$$

$$\left(\mathbf{J}_r^{i+1}\right)^{-1} = \left(\mathbf{J}_r^{i+1,n}\right)^{-1} = \sum_{j \in S^n} \left(\mathbf{J}_r^{j+1}\right)^{-1} \, q^{n,j} \qquad \forall \quad n \in T \tag{25}$$

$$\mathbf{A}_r^{i+1} = \mathbf{A}_r^{i+1,n} = \sum_{j \in S^n} \mathbf{A}_r^{i+1} \, q^{n,j} \qquad \forall \quad n \in T \tag{26}$$

$$\mathbf{B}_r^{i+1} = \mathbf{B}_r^{i+1,n} = \sum_{j \in S^n} \mathbf{B}_r^{i+1} \, q^{n,j} \qquad \forall \quad n \in T \tag{27}$$

$$u^{i+1} = u^{i+1,n} = \sum_{j \in S^n} u^{i+1} \, q^{n,j} \qquad \forall \quad n \in T \tag{28}$$

In addition to the constraints introduced above, which are directly related to implementing the POD-TPWL reservoir model, care must be taken to embed the linear reservoir model in the overarching scope of MILP framework. Here, we highlight two particularly important features for integration with CCS chain optimization: (i) the relevant time horizon and associated time discretization, and (ii) the pressure connection to the transport infrastructure. We note, however, that additional aspects may need attention depending on the intended use of the reservoir model.

2.3.1. MILP parallel time discretization

State-of-the-art CCS chain optimizations typically require the problem to be time discretized. Typically, analyses either focus on multi-year evolution of the CO_2 infrastructure, or on hourly-discretized, year-long evaluation of design and operation. While the first option usually neglects the dynamics of point-sources, e.g. the operation of a power plant, the second does not fully grasp the slow deployment of large infrastructures over a long time horizon. This duality becomes evident when considering the two extremes of a CO_2 chain: while capture plants have dynamics at the hourly level, storage reservoirs see

changes over years and decades. Therefore, when integrating the reservoir model's constraints into the overall chain optimization problem we propose the use of two parallel time discretizations. The first one is the set T, which we set to a timestep length Δt_T of six months and indexes all the constraints related to the reservoir model (Eq. (15)–(28), plus Eq. (14)). The second one is the set T', which has a shorter timestep length $\Delta t_{T'}$ (one day) and indexes all the other elements of the optimization problem, i.e. the operations of CO_2 capture, transport and conditioning.

The two time discretizations are connected to each other assuming that the injection rate in the reservoir u^n is the average of the CO_2 mass flow $\dot{m}^{n'}$ (expressed in volumetric terms) that the CCS infrastructure transports to the storage throughout all the timesteps n' in the corresponding timestep n (we define this subset of the set T' as $T'^{\,n}$):

$$u^n = \sum_{n' \in T'^{\,n}} \frac{\dot{m}^{n'} / \rho^{\text{surf}}_{CO_2}}{|T'^{\,n}| \; \Delta t_{T'}} \qquad \forall \quad n \in T \tag{29}$$

where $|T'^{\,n}|$ is the number of elements in the set $T'^{\,n}$, i.e. it is the number of days in the nth six-month period. In a similar way, whenever the reservoir variables are used in a constraint indexed with $T'^{\,n}$, the corresponding index n is used.

The benefit of having two separate time discretizations is twofold. Firstly, it allows to compute the reservoir states with a timescale that fits the dynamics of the CO_2 flow in a reservoir (which can be in the order of months or years) without compromising the time resolution of the CO_2 infrastructure's operations, where hourly or daily resolutions are needed. Secondly, it limits the computational impact of the reservoir MILP model, as the corresponding variables and constraints are specified only $|T| \ll |T'|$ times.

2.3.2. Connecting the storage model with transport infrastructure

To reconcile the reservoir model to the full chain, we need to link the reservoir pressure to the transportation pressure. With the POD-TPWL model we can obtain the pressure of each grid block in the reservoir over time. Extracting the pressure in the well's grid cell p_{well}, we can approximate the BHP (p_{bh}) by inverting the Peaceman inflow relation if the well perforates a single cell in the reservoir:

$$p^n_{\text{bh}} = p^n_{\text{well}} + \frac{u^n}{\lambda \; WI \; C_{CO_2} \; b_{CO_2}} \qquad \forall \quad n \in T \tag{30}$$

where λ is the total mobility of the reservoir cell, WI is the well index, C_{CO_2} is the fraction of CO_2 in the injected flow (equal to 1 in this case as, for simplicity, we consider injecting pure CO_2), and b_{CO_2} is the ratio between the density of CO_2 at surface and reservoir pressure. We highlight that, for simplicity, we calculate λ using the average value obtained during the training run as the saturation in the well cell. Moreover, we compute the density at reservoir conditions assuming that the pressure is the average of p_{well} during the training run. The BHP depth is given at the centre of the corresponding cell.

To ensure the safety of CO_2 storage, the BHP should not reach the fracture pressure of the reservoir's caprock $p_{\max}$. Therefore, we set the following constraint, which provides a safety margin of 10% from the caprock fracture:

$$p^n_{\text{bh}} \leq 0.9 \; p_{\max} \qquad \forall \quad n \in T \tag{31}$$

To connect the evolution of the pressure of the reservoir with the rest of the CCS infrastructure, we estimate the wellhead pressure p_{wh} as (Ringrose, 2020)

$$p^n_{\text{wh}} = p^n_{\text{bh}} - \rho^{\text{surf}}_{CO_2} g \Delta h \qquad \forall \quad n \in T \tag{32}$$

where g is the gravity constant and Δh is the vertical distance between the surface and the end of the injection well. Assuming that the wellhead system is connected to the pumping system through a pipeline (as for example in Northern Lights and Porthos initiatives), maintaining the flow requires the pumping system to overcome a pressure loss Δp. Therefore, the outlet pressure at the pumping system is given by

$$p^n_{\text{pump, out}} = p^n_{\text{wh}} + \Delta p \qquad \forall \quad n \in T \tag{33}$$

where we have simplified the problem considering no interdependency between wellhead pressure and flow. Consequently, we can calculate the power consumption of the pumping system $W^{n'}_{\text{pump}}$ given a certain inlet pressure $p_{\text{pump, in}}$:

$$W^{n'}_{\text{pump}} = \dot{m}^{n'} \, v \, \frac{p^n_{\text{pump, out}} - p_{\text{pump, in}}}{\eta_{\text{pump}}} \qquad \forall \quad n' \in T' \tag{34}$$

where v is the specific volume of CO_2 and η_{pump} is the efficiency of the pump. Eq. (34) is clearly non-linear. To tackle this issue, we linearized the equation fitting it with a plane in the variables $\dot{m}$ and $p_{\text{pump, out}}$ (see Supplementary Information for more details).

3. Performance of the MILP reservoir model

The MILP reservoir model is validated by comparing the reservoir pressure computed by the reduced-order model and the full scale model. As a reservoir, we considered a saline aquifer operated with realistic injection values and profiles; additional information on the reservoir properties and modelling details can be found in the Appendix. We note that the proposed model can theoretically be used for any type of underground porous formations; however, we have here developed and tuned the model using a saline aquifer exclusively (which provides the largest CO_2 storage availability). Five full-scale simulations of different injections of CO_2 into a saline aquifer with six-month timesteps were carried out and used to build the snapshot matrix described in the Methods section. It is important to note that these simulations are independent of the validation simulations discussed later. Of the training simulations, only one was used to derive the Jacobians for the linear model, which can therefore be regarded as the main training simulation. The main full-scale training simulation starts with an injection rate of approximately 2400 tCO_2/day, increases monotonically to around 3700 tCO_2/day after 2.5 years and remains steady afterwards (see Fig. 2). While we show and benchmark the model along a 5-year period, the training simulations were run with a 30-year period so as to enlarge the pool of snapshots available to the model. The 5-year period was chosen to keep the validation simple and clear.

The linear model's ability to predict the evolution of the pressure in the well's cell was evaluated using four additional distinct injection profiles: (i) a scenario starting and ending at the same injection rates as the training data but with a faster ramp-up (min-to-max in 0.5 years), (ii) a scenario remaining within the minimum and maximum injection rates of the training data but fluctuating with a seasonal pattern, (iii) a scenario with a structurally lower injection rate, and (iv) a scenario with a structurally higher injection rate and a faster ramp-up. More information about the scenarios are reported in Table 1.

As shown in Figs. 2 and 3, the linear model demonstrates a good predictive performance, capturing both the absolute value of the (well) pressure and its evolution, with deviations typically within ±5% of the full-scale simulation results for all tested injection profiles. Overall, the linear model is very accurate for injection rates within the bounds of the training data, as observed in the "faster ramp-up" and "variable injection" scenarios. For the "variable injection" case, minor discrepancies are noted at the beginning of the operations; afterwards, the model computed the pressure in good agreement with the full-scale simulations. However, it fails to capture the small variability in pressure induced by changes in injection rate while retaining the gradual build up. Moreover, the linear model also performs well for the scenarios involving structurally higher or lower injection rates than those in the training data, with values and profiles reasonably close to the full model. It is worth noting that in these cases, the linear

Table 1
Summary of the main injection parameters for the training run and scenarios.

Simulation	Initial injection [t/day]	Final injection [t/day]	Cum. Injection [Mt]	Ramp-up time [y]
Main training run	2403	3713	6.0	1.5
Faster ramp-up	2403	3713	6.0	0.5
Variable injection	2840	2840	5.8	0.5
Lower injection	655	2840	3.7	2
Higher injection	2840	4581	7.7	0.5

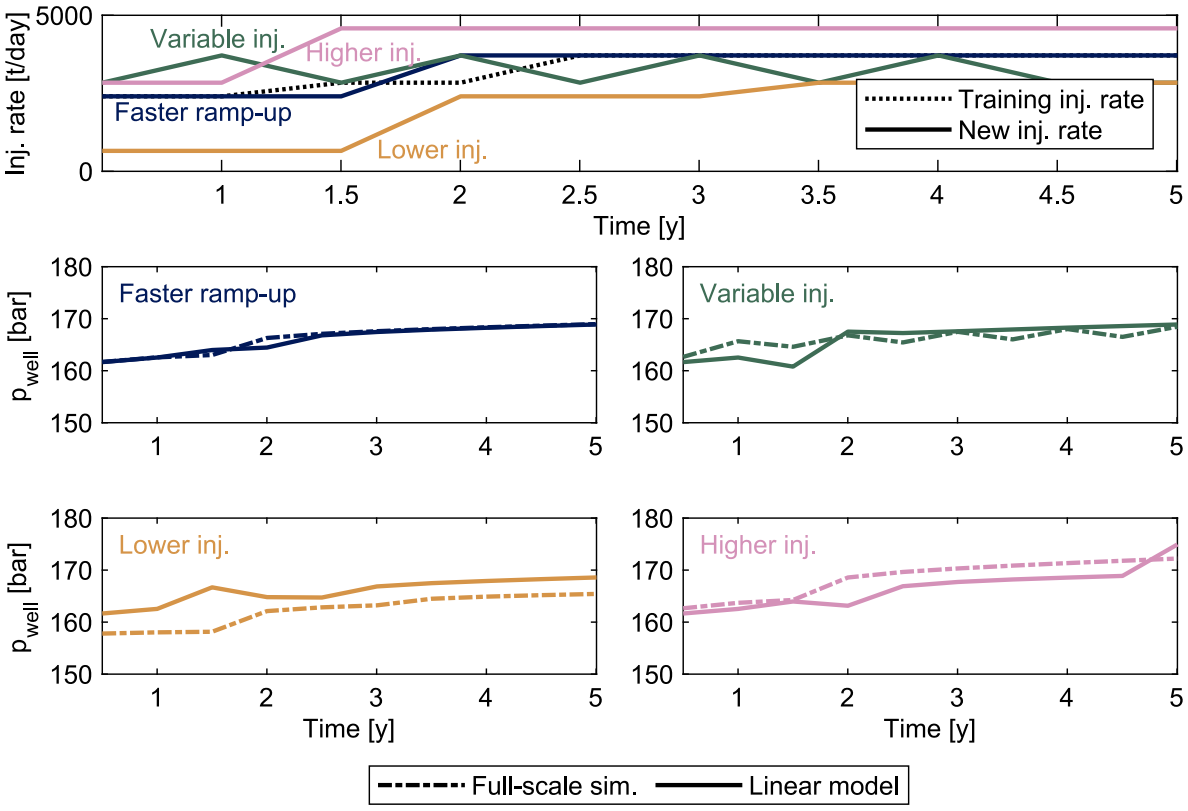


Fig. 2. Top figure: overview of the four injection rate profiles and the training injection rate (t/day over years). Bottom: Time evolution of the well cell pressure under the respective injection rates illustrated above. The linear model (continuous line) is compared with full-scale simulations performed with MRST (dashed line).

model tends to underestimate or overestimate the well cell pressure, respectively to the higher or lower injection rate. This behaviour arises from the model's reliance on a first-order Taylor expansion around the training data, which limits its ability to deviate significantly from the training run's output.

In terms of computational performance, the linear model achieves an average speed-up of approximately 500–1800 times depending on the injection scenario compared to full-scale simulations (see Table 2). Next to the linearization of the governing equations, the driver of the speed-up is the reduction of the dimensionality of the problem. Through POD, this is reduced to 72 ($l_p = 71$ pressure modes and $l_{\text{sat}} = 1$ saturation mode) compared to the full-scale system, which requires 13754 dimensions (two dimensions per grid block). Overall, the dimensions are reduced by more than 99%. It is worth noting that the speed-up obtained in this work is around one order of magnitude higher than the one of Jin and Durlofsky (2018). While a direct comparison cannot be made due to the different full-scale simulations, it is very likely that the faster computation is achieved thanks to the simplifications introduced in the metrics of the TPWL and the restriction of its search region. In addition, we use proportionally less modes, as we are not interested in the temporal evolution of the saturation field.

It is important to note that, since isolating the CO_2 storage linear model within an MILP problem is not feasible, we define its computational time as the sum of the following steps: determining the point of linearization, solving Eq. (14) and rewriting the states in the original basis for all simulation timesteps. This value serves as a reference only, as the computation time of full-scale simulations is influenced

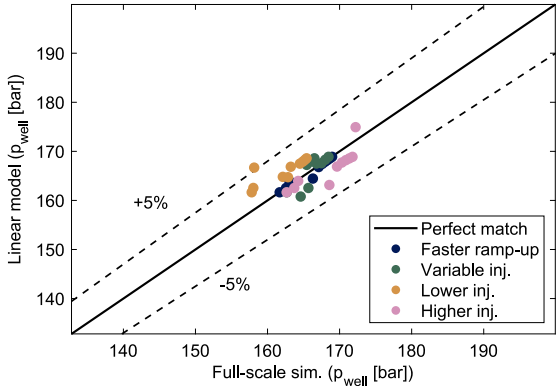


Fig. 3. Parity plot showing the predictive ability of the linear model compared to a full-scale simulation. Each point represents the pressure in the well's cell in each timestep for the respective injection scenario.

Table 2
Computation time of the full-scale simulation and linear model across different injection scenarios.

Scenario	Full-scale simulation [s]	Linear model [s]	Speed-up
Faster ramp-up	5.43	0.0104	522
Variable injection	4.68	0.0045	1040
Lower injection	4.49	0.0044	1020
Higher injection	5.58	0.0030	1860

Table 3
Overview of the main optimization variables, with their time discretization and additional characteristics across the full chain.

Chain element	Optimization variables	Time discretization	Additional characteristics
Emitters with CC	Size of PCC; Flow rate of CO_2 treated by PCC	– ; Day	Daily emission profiles from real-world plants
Pipelines	Size	–	Transport in dense phase
Pumping system	Power consumption; Size	Day; –	Power consumption as a function of mass flow rate and pressure difference required
Storage	Injection rate; Pressure field	Day; 6 months	Saline aquifer

by various factors such as the discretization of the grid, the number of timesteps and the complexity of the full-scale simulation.

4. Optimizing the full CCS chain including reservoir modelling: An illustrative case study

Here, we showcase the integration of the proposed reservoir model into the optimization of the CCS chain using an illustrative case study, where the latter purely serves the purposes of demonstrating how the reservoir model can be used in MILP optimization. The case study considers a typical CCS chain problem: the decarbonization of an industrial cluster comprising a waste-to-energy (WtE) facility and two cement plants that are located onshore at a moderate distance from the offshore reservoir. The case study and its topology are illustrated in Fig. 4. As for the validation, we considered a five-year period with daily resolution for all components excepts the reservoir, which is again modelled with a six-month resolution. We chose a daily resolution because the considered industrial processes have a rather continuous operation, given the nature of their service (waste incineration) or the features of their reactors (high-temperature cement production). Each industrial plant can adopt post-combustion carbon capture (PCC) to avoid CO_2 emissions from entering the atmosphere. To incentivize industrial actors to mitigate their emissions, we considered a scenario with progressively increasing CO_2 tax over time. When PCC is implemented, the captured CO_2 is transported via pipeline in the dense (supercritical) phase to a central onshore collection point before injection. Here, CO_2 is further pressurized to the required storage conditions, and injected via an offshore pipeline in the saline aquifer. An overview of the main optimization variables is reported in Table 3, and further assumptions underlying the case study are reported in the Supplementary Information. The optimization problem was formulated using the peer-reviewed, open-source MILP energy system optimization software AdOpT-NET0 (Wiegner et al., 2025), which is implemented using the Pyomo modelling language (Hart et al., 2011). The interested user can find there and within the referenced scientific papers all additional information regarding the model and its specific formulation. The MILP optimization problem has 220532 continuous variables, 94 binaries and solves in 94.39 s; it was solved using Gurobi version 11.0.2 with an optimality gap of 2%, executed on a 12th Gen Intel Core i7-1255U processor (10 cores, 12 threads).

Fig. 5(a) illustrates the overall industrial cluster's emission profile over time, with emissions captured and emitted to the atmosphere. Under the case study's conditions, the cost-optimal strategy for all point sources in the cluster involves deploying PCC units to mitigate a portion of their emissions. However, the cluster does not achieve full-capacity carbon capture immediately: This delay is attributable to the WtE facility, specifically to the lower CO_2 concentration compared to the cement plant, which results in higher operational costs. As a result, when the CO_2 tax is lower, it is not cost-efficient to reduce emissions in the WtE plant. In addition, the CO_2 capture profile reveals a seasonal pattern, which follows the operation of the units (considered as exogenous in this case study). The resulting seasonal variation in the capture profile is of particular interest in this study, as it is effectively enabled by a varying injection profile in the reservoir. Indeed, the captured CO_2 leads to an optimal average injection rate over six-month timesteps that fluctuates seasonally between 4100 t/day and 3700 t/day (Fig. 5(b)). This average injection rate exhibits a bi-modal trend, reflecting the variability in the emission profile of the cement plants. The resulting injection rate dictates the evolution of the BHP in the reservoir, which increases over time but remains well below the caprock fracture pressure, as ensured by the constraint in Eq. (31).

On the other hand, the evolution of pressure in the reservoir impacts both the sizing and the energy consumption of the pumping system, which connects the onshore pipeline to the saline aquifer via an offshore pipeline. As reservoir pressure increases over time, the specific energy consumption of the pump also increases. This increase is in the order of 20-40% throughout the analysed time frame. While the pump station consumes only 1–1.5 kWh/tCO_2 with limited impact on the overall energy consumption of the chain (note that the injection pump compresses CO_2 in the dense phase), variations are likely to affect the size and pump type required to accommodate the pressure fluctuations.

5. Discussion

The model developed in this work bridges the gap between high-level optimization required to design the full CCS chain and the reservoir's dynamics and operational constraints. This approach ensures that the volumes of CO_2 determined to be optimal for capture and transport within the CCS chain are also operationally feasible for the reservoir. As first approximation, we constrained the bottom-hole pressure in the reservoir to remain below the caprock fracture pressure. We used the model in an applied example to explore how a varying injection rate can accommodate variations in CO_2 captured at an industrial cluster. It is important to note that implementing this model does not replace comprehensive full-scale reservoir simulations, which are essential for achieving precise evaluations and making final decisions, but opens up opportunities for improved chain configurations and for direct communications between reservoir models and CO_2 chain models. In the following paragraphs, we discuss the model complexity and limitations, its portability to different MILP frameworks, additional potential applications and its economic implications.

Computational complexity and limitations. The integration of the reservoir model within an MILP optimization framework for CCS chain introduces extra computational complexity to the problem, which impacts its solving time. While the solving time of the case study is still very low (between one and two minutes), the increase compared to solving the same case study without the reservoir model is in the order of 10^2. Consequently, integrating the model into very large MILP

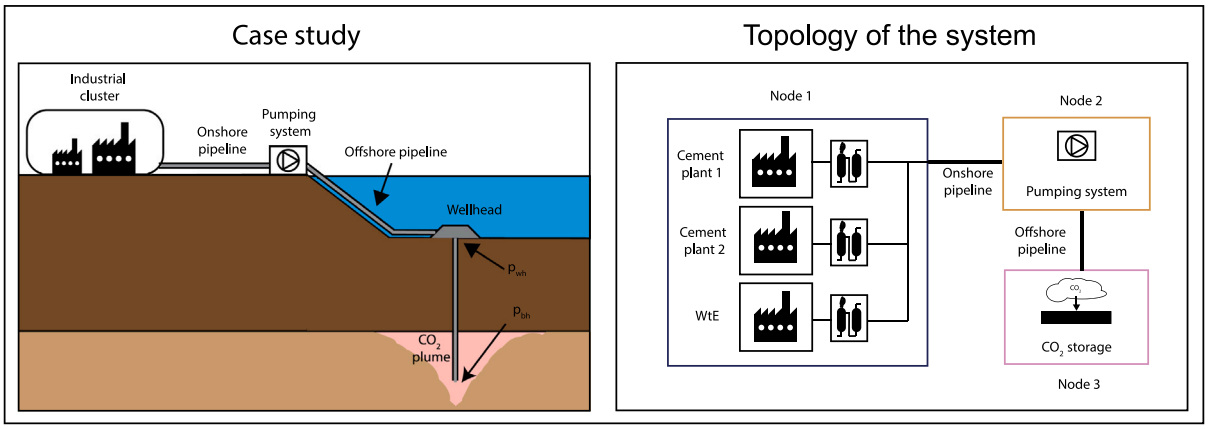


Fig. 4. Schematic representation of the illustrative case study and the relative topology. Left: graphical illustration of the case study, with emphasis on the storage components; right: equivalent case study network.

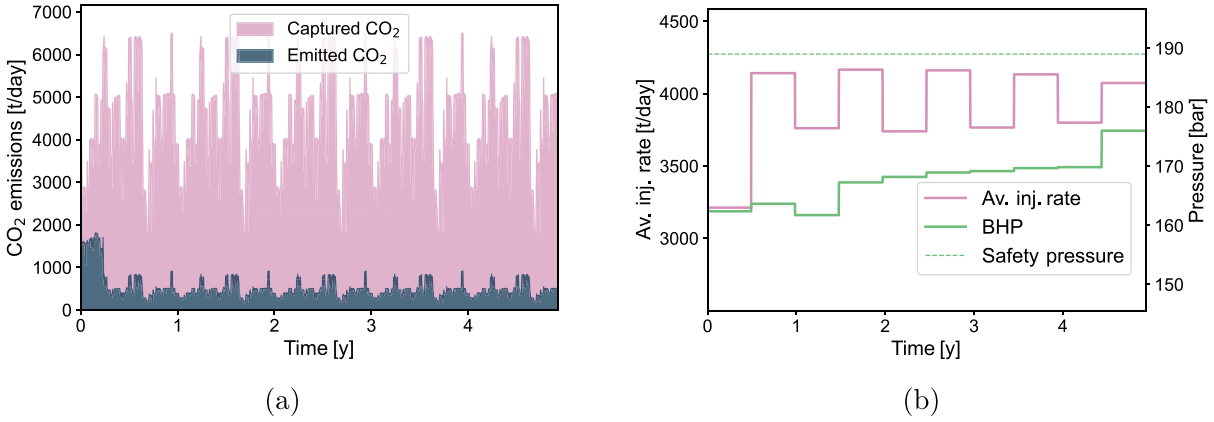


Fig. 5. Evolution of the emission profile of the industrial cluster and impact of resulting optimal CO_2 injection rate on the reservoir's pressure. (a) Emission profile of the overall industrial cluster. (b) Injection rate in the reservoir averaged over six-month periods and evolution of the bottom-hole pressure over time. The figure also shows the maximum allowed pressure to ensure the storage safety (90% of the caprock fracture pressure p_{max}).

problems might present challenges, especially in sector-coupled studies such as electricity and gas infrastructure co-optimization. The number of variables necessary for the MILP reservoir model in the optimization problem is influenced by three primary parameters: (i) the number of elements in the search group (S^n), (ii) the number of modes, and (iii) the number of timesteps. In this study, the number of elements in S^n is fixed at three, representing the minimum feasible value, and thus cannot be further reduced. The number of modes directly relates to the number of variables and constraints, making it an important factor in determining the computational burden. Reducing the number of modes could therefore significantly alleviate the computational complexity. However, the minimum number of modes is approximately proportional to the number of gridblocks in the full-scale reservoir simulation. Therefore, reducing the number of gridblocks could decrease the number of modes, but this must be done cautiously to avoid weakening the full-scale model's ability to represent the reservoir physics accurately. One option to decrease the number of gridblocks while preserving the model's capabilities could be using hybrid vertical equilibrium models (Møyner and Nilsen, 2019). Here, the rapid vertical segregation due to the large density difference between water and CO_2 is exploited to reduce the number of degrees of freedom in large parts of the simulation domain. Another potential strategy for mitigating computational complexity involves increasing the length of the reduced timestep (e.g., transitioning from 6-month to 1-year timesteps). The number of variables and constraints scales linearly with the number of reduced timesteps, enabling a proportional reduction in these quantities. However, it is important to note that (i) such reductions may not lead to a corresponding linear decrease in computational time and (ii) it may hinder the quality of the model output.

Model portability. The model we present here is, in principle, portable to any MILP optimization framework where it is possible to add user-defined objects. However, two factors need to be taken into account. First, tuning the model to a specific injection site requires having a full-scale reservoir simulator; this should also come with access to the code to extract the Jacobians needed for the linearization process. While this is feasible using open-source software such as MRST (a detailed explanation of how the Jacobians are derived in MRST is provided in the Supplementary Information), it demands familiarity with the principles of full-scale reservoir simulations. Second, the MILP implementation requires the formulation of two distinct time discretizations for different components of the MILP problem, which might not be trivial in existing MILP tools.

Potential applications of the model. The model offers potential for additional applications. We applied the model to a simple case study. However, the model adds greater value when incorporated in larger-scale, long-term CCS chain planning. In this context, the timescale of the chain aligns better with the timescale of the reservoir dynamics; moreover, large-scale planning involves a high degree of uncertainty (on both the magnitude and timeline of injection) and, therefore, knowing how the reservoir responds to changes is extremely relevant. To further improve the relevance of the model, the coupling of the evolution of reservoir pressure and the transport capacity of the pipeline system connecting to the wellhead could be implemented. The transport capacity of such a pipeline depends on the wellhead pressure,

which in turn influences the overall capacity of the CCS infrastructure to manage the anticipated volume of captured CO_2. Omitting this phenomenon in the optimization of the CCS chain may lead to solutions that are in fact unfeasible. However, this integration is challenging due to the nonlinear relationship between pipeline transport capacity and wellhead pressure (Anon, 2019). Future work would need to explore appropriate strategies for linearizing this dependency to enable its incorporation into the optimization framework.

For another potential application of the model, the optimization problem could be reformulated to prevent CO_2 from reaching areas in the reservoir associated with higher leakage risks. This could be achieved by imposing constraints on the CO_2 saturation in specific reservoir cells. While POD-TPWL models have previously been used to minimize CO_2 accumulation in target areas of a reservoir (Jin and Durlofsky, 2018), the accuracy of the linear model in such applications would require careful validation.

Economic implications: Looking beyond the illustrative case study developed in this work, the economic implications of the reservoir model might be two-fold. First, the model allows to avoid CCS chain designs that might be not compatible with the reservoir dynamics; as the reservoir is a major bottleneck of the chain, accounting for these dynamics directly in the chain optimization might unlock new optimal configurations (i.e. lower costs) that are difficult to identify when using separate simulations. Second, including a realistic model of CO_2 storage enables better costs estimations of the logistic part of the chain (design pressure of pipelines, size of compression/pumping systems etc.). However, these costs are highly project-dependent, and a thorough economic assessment of a real-world project is beyond the scope of this work.

6. Conclusion

In this work, we developed an MILP formulation that integrates reservoir modelling with the optimization of the entire CCS chain. To do so, we used a POD-TPWL model to tackle the high-dimensionality and nonlinearity of the governing equations of CO_2 injection in a reservoir. Specifically, with proper orthogonal decomposition, the system is projected onto an appropriate subspace, reducing the dimensions of the system by more than 99% while retaining most of the information. The linearization of the governing equations is achieved using the trajectory piecewise linearization technique, which applies a first-order Taylor expansion around a training solution that is dynamically selected minimizing a suitable metric. Differently from other POD-TPWL models, we modified the metric to (i) make it fully linear and (ii) limit the computational burden of the model and boost its application to large-scale MILP. We then developed the necessary MILP constraints to integrate the POD-TPWL model in the optimization of the CCS chain with MILP.

The model was compared with full-scale reservoir simulations under different injection scenarios. The model showed good accuracy in predicting the evolution of the well cell pressure, with deviations typically in the range of ±5%. This is achieved with a speed-up in the range of 500–1800×.

We then developed an illustrative case study, where we optimized the design of the CCS chain to decarbonize an industrial cluster consisting of cement and waste-to-energy plants. We showed how time-varying CO_2 capture at the point-sources plants can be accommodated with time-varying injection rates in the reservoir (while keeping injection continuous). In this context, integrating the reservoir model ensured that, within its accuracy, the pressure in the reservoir is kept within safety limits given the optimal volumes of CO_2 captured and stored.

Overall, we demonstrated that reservoir simulations can be incorporated into the optimization of the CCS chain. This comes, however, with some limitations. In particular, the model adds complexity to the optimization problem, which means that integrating it into very large case studies might be challenging. The model also requires familiarity with both reservoir simulations and MILP modelling, making it not necessarily trivial to implement.

CRediT authorship contribution statement

Luca Bertoni: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Olav Møyner:** Writing – review & editing, Software, Methodology. **Jan Wiegner:** Writing – review & editing, Software, Methodology. **Matteo Gazzani:** Writing – review & editing, Visualization, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT and Gemini in order to explore possible improvements of the authors' text. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.compchemeng.2025.109293.

Data availability

Data will be made available on request.

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